Dimensions and Units

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Measurements

Used to describe natural phenomena

Each measurement is associated with a physical quantity

Need defined standards

Characteristics of standards for measurements

- Readily accessible
- Possess some property that can be measured reliably
- Must yield the same results when used by anyone anywhere
- Cannot change with time

Basic Physical Quantities

- **There are certain *basic physical quantities* which are *defined through the way they are measured.* Such a procedure for defining the basic physical quantities is called *operational procedure.***
- **Other physical quantities are *derived* from the basic ones *through the mathematical relations that relate them.***
- **The choice of the basic quantities depends to some extent on the system of units used.**
- **The most widely used system of units now is the SI "Système International".**

Fundamental Quantities and Their Units

Quantity	SI Unit
Length	meter
Mass	kilogram
Time	second
Temperature	Kelvin
Electric Current	Ampere
Luminous Intensity	Candela
Amount of Substance	mole

Basic Quantities and Their Dimension

Dimension has a specific meaning – it denotes the physical nature of a quantity.

Dimensions are often denoted with square brackets.

- Length [L]
- Mass [M]
- Time [T]

Length

Length is the distance between two points in space.

Units

- SI – meter, m

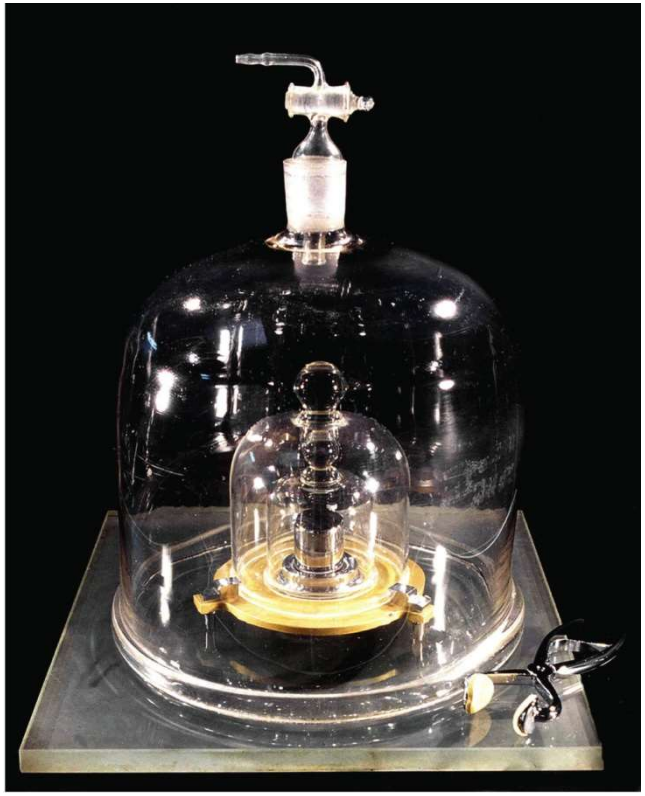
Defined in terms of a meter – the distance traveled by light in a vacuum during a given time

Mass

Units

- SI – kilogram, kg

Defined in terms of a kilogram, based on a specific cylinder kept at the International Bureau of Standards



a

Standard Kilogram

Time

Units

- seconds, s

Defined in terms of the oscillation of radiation from a cesium atom

See Table 1.3 for some approximate time intervals.

US Customary System

Still used in the US, but text will use SI

Quantity	Unit
Length	foot
Mass	slug
Time	second

I. Units and Dimensions of Physical Quantities

Each dimension can have many actual units.

Table 1.5 for the dimensions and units of some derived quantities

TABLE 1.5 *Dimensions and Units of Four Derived Quantities*

Quantity	Area (A)	Volume (V)	Speed (v)	Acceleration (a)
Dimensions	L^2	L^3	L/T	L/T^2
SI units	m^2	m^3	m/s	m/s^2
U.S. customary units	ft^2	ft^3	ft/s	ft/s^2

Example 1:

Determine the dimensions and SI units of the following physical quantities:

- Force
- Pressure
- Potential Energy
- Work
- Power

Solution :

$$[F] = [m][a] = M\ L\ T^{-2}$$

$$[P] = \left[\frac{F}{A}\right] = \frac{M\ L\ T^{-2}}{L^2} = M\ L^{-1}\ T^{-2}$$

$$[W] = [F][x] = M\ L\ T^{-2} L = M\ L^2\ T^{-2}$$

$$[P.E.] = [m][g][h] = M\ L\ T^{-2} L = M\ L^2\ T^{-2}$$

$$[Power] = \left[\frac{W}{t}\right] = \frac{M\ L^2\ T^{-2}}{T} = M\ L^2\ T^{-3}$$

II. Units and Dimensions of Physical Constants

Example 2:

Newton's law of universal gravitation is represented by $F = \frac{GMm}{r^2}$, where *F* is the gravitational force exerted by one small object on another, *M* and *m* are the masses of the objects, and *r* is a distance. What are the SI units of the Universal Gravitational constant "*G*" ?

Solution :

$F = \frac{GMm}{r^2}$, then $G = \frac{Fr^2}{Mm}$

$$[G] = \frac{[F][r]^2}{[M][m]} = \frac{MLT^{-2}L^2}{MM} = M^{-1}T^{-2}L^3$$

Dimensional Analysis

- Technique to check the correctness of an equation or to assist in deriving an equation
- Dimensions (length, mass, time, combinations) can be treated as algebraic quantities.
- Both sides of equation must have the same dimensions.
- Any relationship can be correct only if the dimensions on both sides of the equation are the same.
- Cannot give numerical factors: this is its limitation

III. Check of a Physical Relation

Example 3: Is this equation correct velocity ?

$$v = v_0 + \tfrac{1}{2}at^2.$$

Solution:

Check the dimensions of the terms of the equation

$$\left[\frac{L}{T}\right] \stackrel{?}{=} \left[\frac{L}{T}\right] + \left[\frac{L}{T^2}\right][T^2] = \left[\frac{L}{T}\right] + [L].$$

Dimensions of the terms are not equal ⇒ **Wrong**

Example 4:

For the wave equation,

$y = C_1 \sin(C_2 t - C_3 x + C_4)$,

if *x* and *y* are distances, t is time, then,

determine the dimensions of C_1 , C_2 , C_3 and C_4

Solution:

***L*, T^{-1} , L^{-1} , Dimensionless**

IV. Building a Physical Relation bet. Dependent Parameters

Dimensional Analysis can be used

- to verify the validity of a certain equation from a dimensional point of view as in the previous example
- to deduce the relationship between a certain physical quantity and the physical quantities it depends on.

Example 5:

Use dimensional analysis to derive the relationship between the time of small oscillations T of a simple pendulum and the length l of the pendulum and the acceleration of gravity g.

Solution

The relation between the periodic time *T* and other parameters can be written as

$$T = Km^{\alpha} l^{\beta} g^{\gamma}$$

K is a numerical constant

The dimensions of the sides of the equation is

$$[T] = [\mathrm{M}]^{\alpha} [L]^{\beta} [LT^{-2}]^{\gamma}$$

By Equating the powers of *M, L, T* over both sides

$$0 = \alpha \qquad 0 = \beta + \gamma \qquad 1 = -2\gamma$$

Solution (cont.)

Then,

$$\alpha=0 \qquad \gamma=-\frac{1}{2} \qquad \beta=-\gamma=\frac{1}{2}$$

Then,

$$T=C\sqrt{\frac{L}{g}}$$

https://www.youtube.com/watch?v=-q18cgk9_Bs